Chapter 6-Data rate and channel capacity

Data Rate Limit

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel.

Data rate depends on three factors:

1. *The bandwidth available*
2. *The level of the signals we use*
3. *The quality of the channel (the level of noise)*

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for noiseless channel, another by Shannon for a noisy channel.

Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 \times \text{Bandwith} \times \log_2 \text{Levels}$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and bit rate is the bit rate in bits per second.

Noiseless Channel: Nyquist Bit Rate

$$\text{BitRate} = 2 \times \text{Bandwith} \times \log_2 \text{Levels}$$

According to the formula, we might think that, given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels. Although the idea is theoretically correct, practically there is a limit. When we increase the number of signal levels, we impose a burden on the receiver. If the number of levels in a signal is just 2, the receiver can easily distinguish between a 0 and a 1. If the level of a signal is 64, the receiver must be very sophisticated to distinguish between 64 different levels. In other words, increasing the levels of a signal reduces the reliability of the system.

Example

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What is the maximum bit rate?

Solution

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). What is the maximum bit rate?

Solution

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Example

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L$$

$$\log_2 L = 6.625 \qquad L = 2^{6.625} = 98.7 \text{ levels}$$

We can Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate.
If we have 128 levels, the bit rate is 280 kbps.
If we have 64 levels, the bit rate is 240 kbps.

Shannon Capacity

In reality, we cannot have a noiseless channel

For noisy channel,

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1+\text{SNR})$$

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.

Example

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. What is the channel capacity?

Solution

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example

Let's calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. What is the channel capacity?

Solution

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example

The signal-to-noise ratio is often given in decibels. Assume that SNR_{dB} = 36 and the channel bandwidth is 2 MHz. What is the theoretical channel capacity?

Solution

$$\mathrm{SNR_{dB}} = 10 \log_{10} \mathrm{SNR} \rightarrow \mathrm{SNR} = 10^{\mathrm{SNR_{dB}}/10} \rightarrow \mathrm{SNR} = 10^{3.6} = 3981$$

$$C = B \log_2 (1 + \mathrm{SNR}) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Example

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \rightarrow \quad L = 4$$

Transmission of digital signal

A digital signal, periodic or nonperiodic, is a composite analog signal with frequencies between zero and infinity.

The fundamental question is, how can we send a digital signal from point A to point B?

We can transmit a digital signal by using one of two different approaches:

- *Baseband transmission or*
- *Broadband transmission (using modulation).*

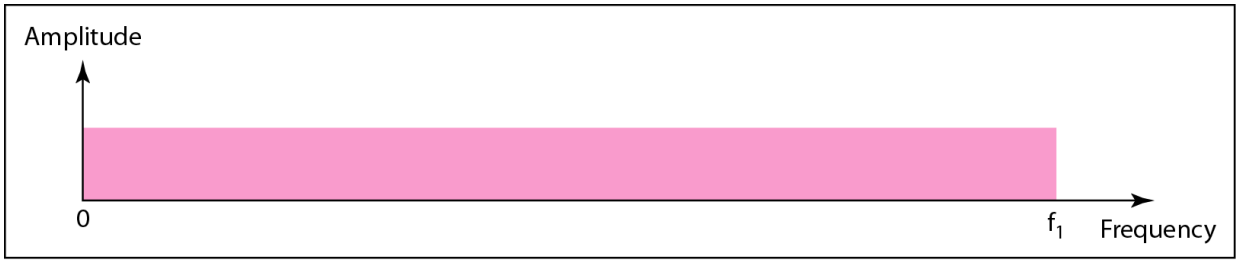
Baseband Transmission

- *Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.*
- *Baseband transmission requires that we have a low-pass channel, a channel with a bandwidth that starts from zero. This is the case if we have a dedicated medium with a bandwidth constituting only one channel.*
- *For example, the entire bandwidth of a cable connecting two computers is one single channel.*
- *As another example, we may connect several computers to a bus, but not allow more than two stations to communicate at a time. Again we have a low-pass channel, and we can use it for baseband communication.*

.

Baseband Transmission

Figure below shows two low-pass channels: one with a narrow bandwidth and the other with a wide bandwidth. We need to remember that a low-pass channel with infinite bandwidth is ideal, but we cannot have such a channel in real life. However, we can get close.



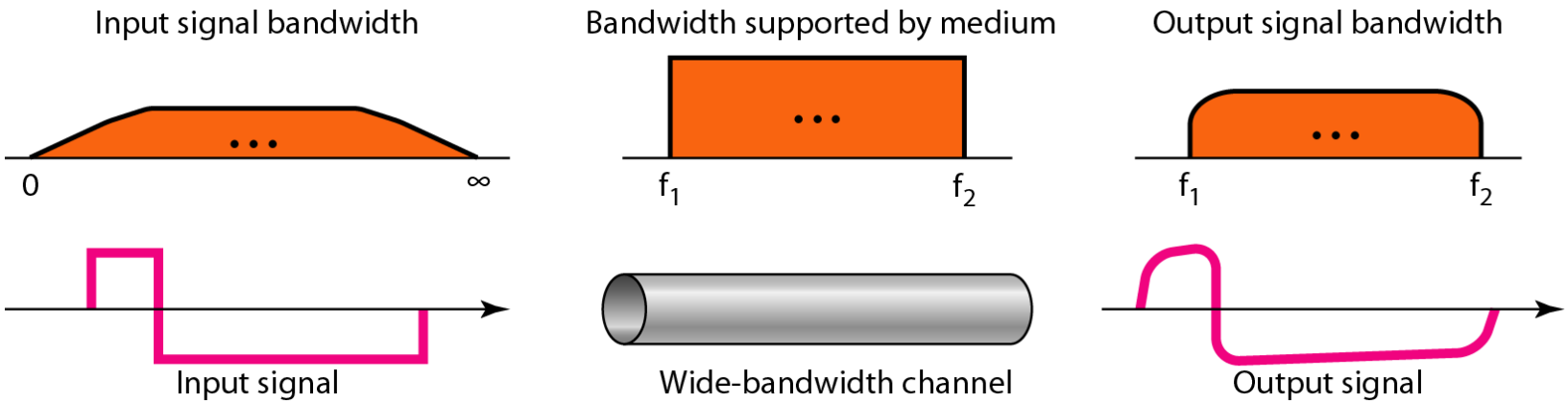
a. Low-pass channel, wide bandwidth

b. Low-pass channel, narrow bandwidth

Baseband transmission using a dedicated medium

- *Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.*
- *If we want to preserve the exact form of a nonperiodic digital signal with vertical segments vertical and horizontal segments horizontal, we need to send the entire spectrum, the continuous range of frequencies between zero and infinity.*
- *This is possible if we have a dedicated medium with an infinite bandwidth between the sender and receiver that preserves the exact amplitude of each component of the composite signal.*
- *Although this may be possible inside a computer (e.g., between CPU and memory), it is not possible between two devices.*
- *Fortunately, the amplitudes of the frequencies at the border of the bandwidth are so small that they can be ignored.*
- *This means that if we have a medium, such as a coaxial cable or fiber optic, with a very wide bandwidth, two stations can communicate by using digital signals with very good accuracy*

Baseband transmission using a dedicated medium



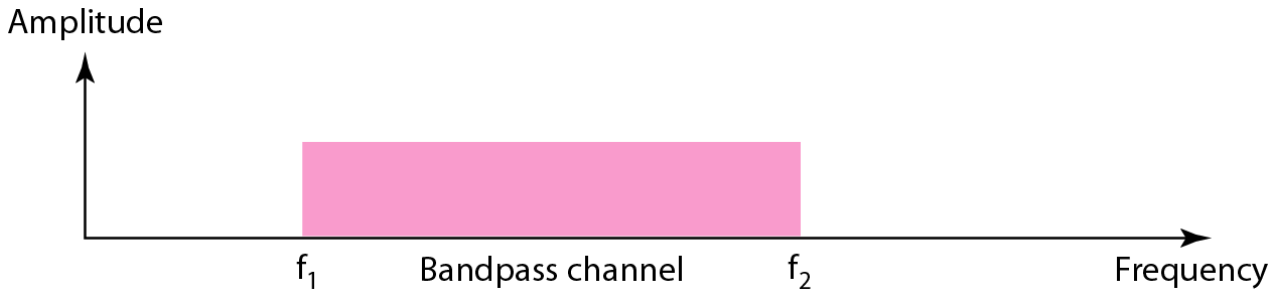
Broadband transmission (Using Modulation)

- *If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.*

- *Broadband transmission or modulation means changing the digital signal to an analog signal for transmission.*

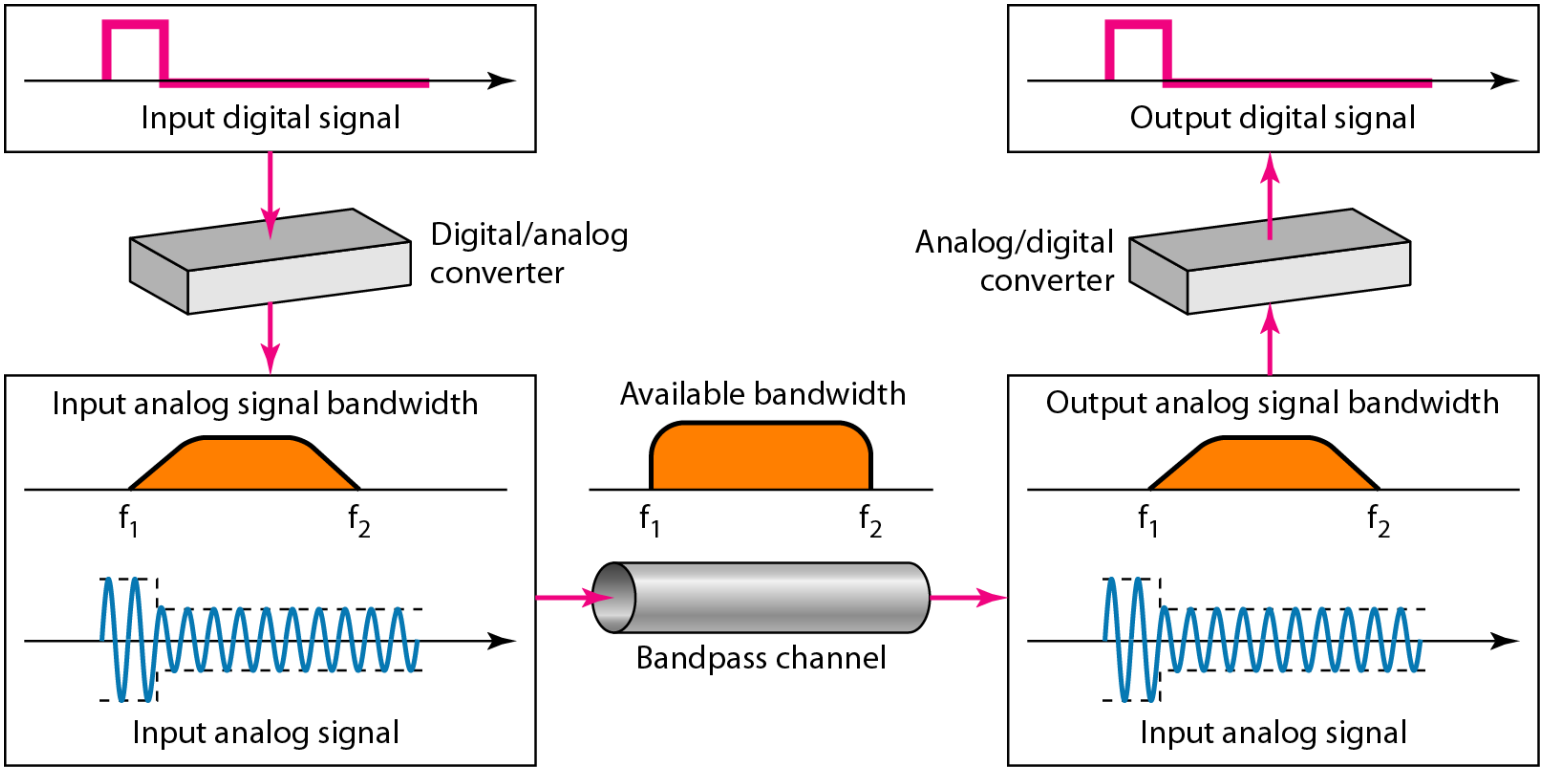
- *Modulation allows us to use a bandpass channel-a channel with a bandwidth that does not start from zero.*

- *This type of channel is more available than a low-pass channel.*



Broadband transmission (Using Modulation)

Modulation of a digital signal for transmission on a bandpass channel is illustrated below



Broadband transmission-Example

An example of broadband transmission using modulation is the sending of computer data through a telephone subscriber line, the line connecting a resident to the central telephone office. These lines are designed to carry voice with a limited bandwidth. The channel is considered a bandpass channel.

We convert the digital signal from the computer to an analog signal, and send the analog signal. We can install two converters to change the digital signal to analog and vice versa at the receiving end. The converter, in this case, is called a modem.